YETI PAW · CABLE HANDLE · RESEARCH BOOK

TWO PIVOT POSITIONS, FOUR MOVEMENTS

The YETI PAW can be built with the cable hooked on in one of two places. The current pivot puts most of the load on the forearm pad. The centered pivot shares it between hand and wrist, closer to a D-ring handle or a wrist cuff. This book runs that comparison on the lateral raise, the bicep curl, the Bayesian curl and the chest fly, and ends with the hardware it led to. The case is for offering both as variants.

V2.0 · 2026-10-17 · CABLE SOURCE: VOLTRA, 45° RACK BRACKET, EXIT 8 IN. UP · FLY: FLUSH-MOUNTED PAIR JUST BELOW SHOULDER HEIGHT

THE BOTTOM LINE

13–17 lb

Current: forearm-led. Of a 20 lb cable, the forearm pad carries 13–17 lb at the hardest point of every movement. The grip can relax.

6–8 lb

Centered: balanced. At the same moment the pad carries 6–8 lb and the hand the rest: between hand and wrist, closer to a D-ring or cuff.

+7–16%

A little more work. The centered pivot's peak effort at the working joint, against the current pivot's. The resistance curve keeps its shape.

MOVEMENT	HARDEST POINT	CURRENT · PAD + HAND	CENTERED · PAD + HAND	CENTERED VS CURRENT, PEAK
Lateral raise	44°	17 + 3 lb	**8 + 12 lb**	+7%
Bicep curl	115°	13 + 7 lb	**6 + 14 lb**	+16%
Bayesian curl	89°	13 + 7 lb	**6 + 14 lb**	+16%
Cable chest fly	31°	17 + 3 lb	**8 + 12 lb**	+7%

Reading the table: 20 lb on the cable, at the current pivot's hardest point; both pivots are compared at that same moment. The last column compares each pivot's own peak. The raise and the fly (and the two curls) split the same way because at the hardest point the cable pulls nearly square to the limb in each pair.

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THE SETUP

A Voltra on a 45° rack bracket at the foot of an upright, cable exit 8 in. off the floor, for the raise and both curls; a pair flush-mounted on uprights just below shoulder height for the fly. One lifter, 6′5″, measured.

THE ANGLES

Every comparison is made at the current pivot's hardest point, the angle where its effort peaks. Each study states its own zero.

THE COLORS

Red is the centered pivot and the forearm pad; gray is the current pivot; **blue** is the hand and the D-handle; dotted black is a wrist cuff.

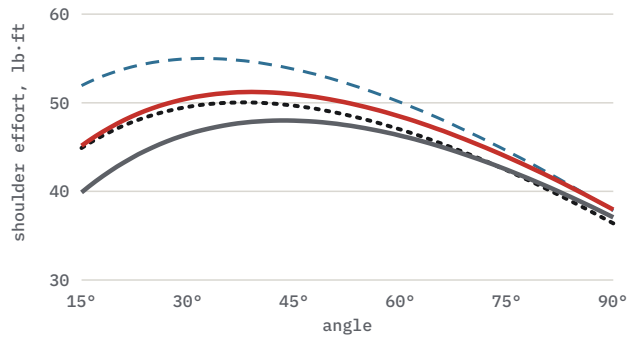
How sure are we? Confident on the numbers for the lateral raise, which rests on measurements, and on the direction for the curls and the fly, whose stances are assumed. Not yet on feel. One physics model of one measured lifter (6′5″), worked twice from scratch, stable when the body and cable-exit guesses move by several inches.

What it can’t tell us: how either handle feels in the hand, or how lifters of other sizes compare. Next step: 3–5 lifters try both variants back to back, with a wrist cuff for reference.

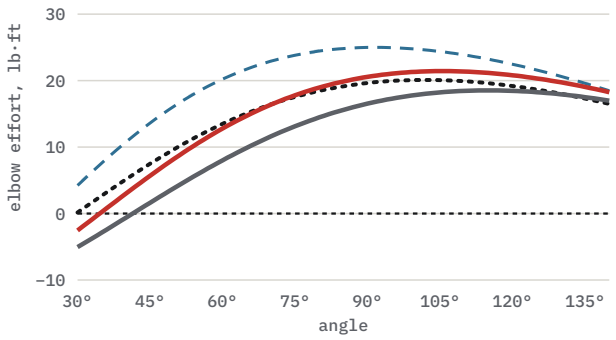
RESISTANCE CURVES

Effort at the working joint through each movement, for 20 lb on the cable. Higher means harder at that angle (lb·ft: pounds times feet of leverage); below zero the cable is helping the limb along.

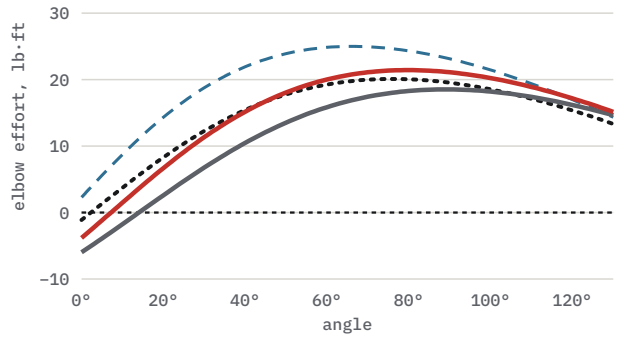
LATERAL RAISE



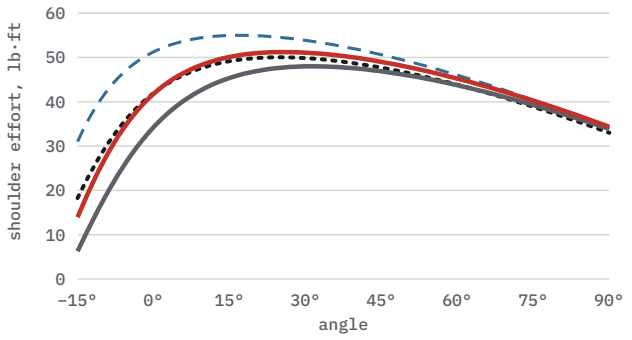
BICEP CURL



BAYESIAN CURL



CABLE CHEST FLY



Wrist cuff · D-handle · YETI PAW · current · YETI PAW · centered pivot

PEAK EFFORT, LB·FT	CURRENT	CENTERED	WRIST CUFF	D-HANDLE	CENTERED VS CURRENT
Lateral raise	48.0 at 44°	**51.2 at 39°**	50.0 at 38°	55.0 at 32°	+7%
Bicep curl	18.5 at 115°	**21.4 at 106°**	20.1 at 102°	25.0 at 92°	+16%
Bayesian curl	18.5 at 89°	**21.4 at 80°**	20.1 at 76°	25.0 at 66°	+16%
Cable chest fly	48.0 at 31°	**51.2 at 26°**	50.0 at 25°	55.0 at 17°	+7%

Reading across: on every movement the four handles draw the same overall shape. The centered pivot peaks 5–9° earlier than the current one and 7–16% higher, within 1–4° and 2–7% of the wrist cuff's peak. The D-handle peaks earliest and highest every time. Each study's “Same path as a cuff” page has the curve at full size; its tables round to whole lb·ft, and every percentage is taken from the unrounded values.

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.0 · 2026-10-17

YETI PAW · CABLE LATERAL RAISE · DESIGN STUDY

LATERAL RAISE: TWO WAYS TO CARRY THE LOAD

The YETI PAW can be built with the cable hooked on in one of two places. Both work, and they suit different lifters. The current pivot puts most of the load on the forearm pad. A centered pivot hooks on near the wrist and shares the load more evenly between hand and wrist, closer to how a D-ring handle or a wrist cuff feels. The case here is for offering both as variants.

THE BOTTOM LINE

17 + 3 lb

Current: forearm-led. At the hardest point of the movement the forearm pad carries 17 lb of a 20 lb cable and the hand 3 lb. The grip can relax and the forearm does the work.

8 + 12 lb

Centered: balanced. The same moment: 8 lb on the pad, 12 lb in the hand. The load sits between hand and wrist instead of mostly on the forearm, closer to a D-ring or cuff.

Plates only

Small change to make. Same handle, pad, yoke and hardware. Only the side plates change: the yoke hole moves to the middle of the handle and pad. Now drawn and checked: at least as strong as the current plate (100 lb cable: static 4.4, fatigue 1.6 vs 1.5), with the yoke free to follow the cable through all four movements studied.

What doesn't change: the exercise itself. The cable follows almost the same path with either pivot; the effort it takes differs by at most 14% at the very start of the movement and 2% at the top. The choice is mainly about where the lifter feels the load.

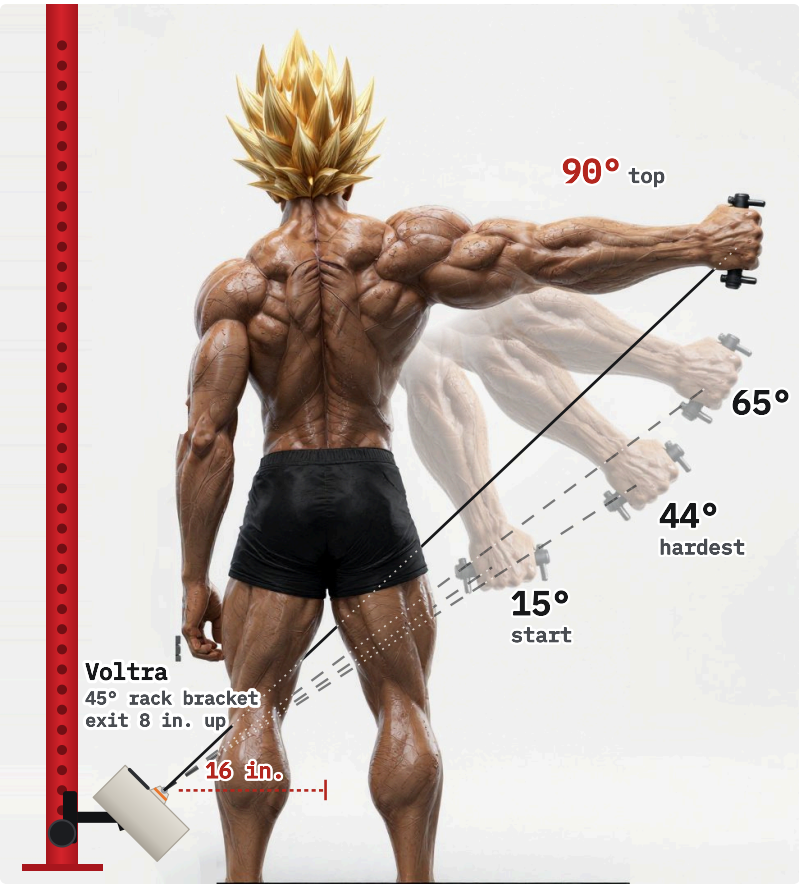
How sure are we? Confident on the numbers, not yet on feel. They come from a physics model of one real lifter (6′5″, measured), were worked out twice from scratch and matched within 1%, and hold up when the body and cable-exit guesses move by several inches.

What it can't tell us: how either handle feels in the hand, or how lifters of other sizes compare. Nobody has used a centered-pivot prototype yet.

Next step: centered-pivot side plates are printing now. Have 3–5 lifters try both variants back to back, with a wrist cuff for reference.

THE TEST MOVEMENT

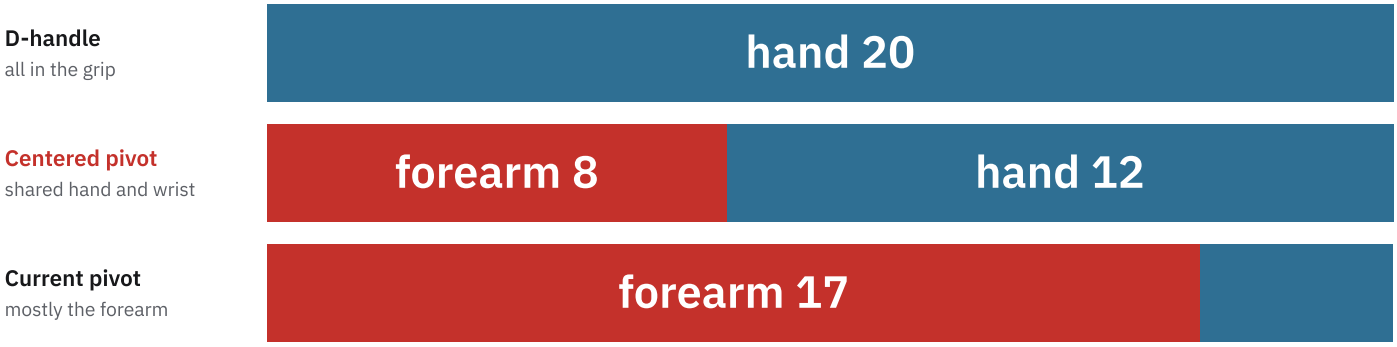
The comparison needs one movement that swings the arm through a wide range of angles against the cable, so a cable lateral raise is used as the yardstick. It is a measuring stick, not the point: the same shift between hand and forearm shows up in any movement where the cable pulls across the forearm, though the exact pounds depend on the angle. Seen from behind, the cable runs from a Voltra on a 45° rack bracket beside the left foot (cable exit 8 in. up), across the front of the body, to the right hand; the arm rises from 15° to 90°.



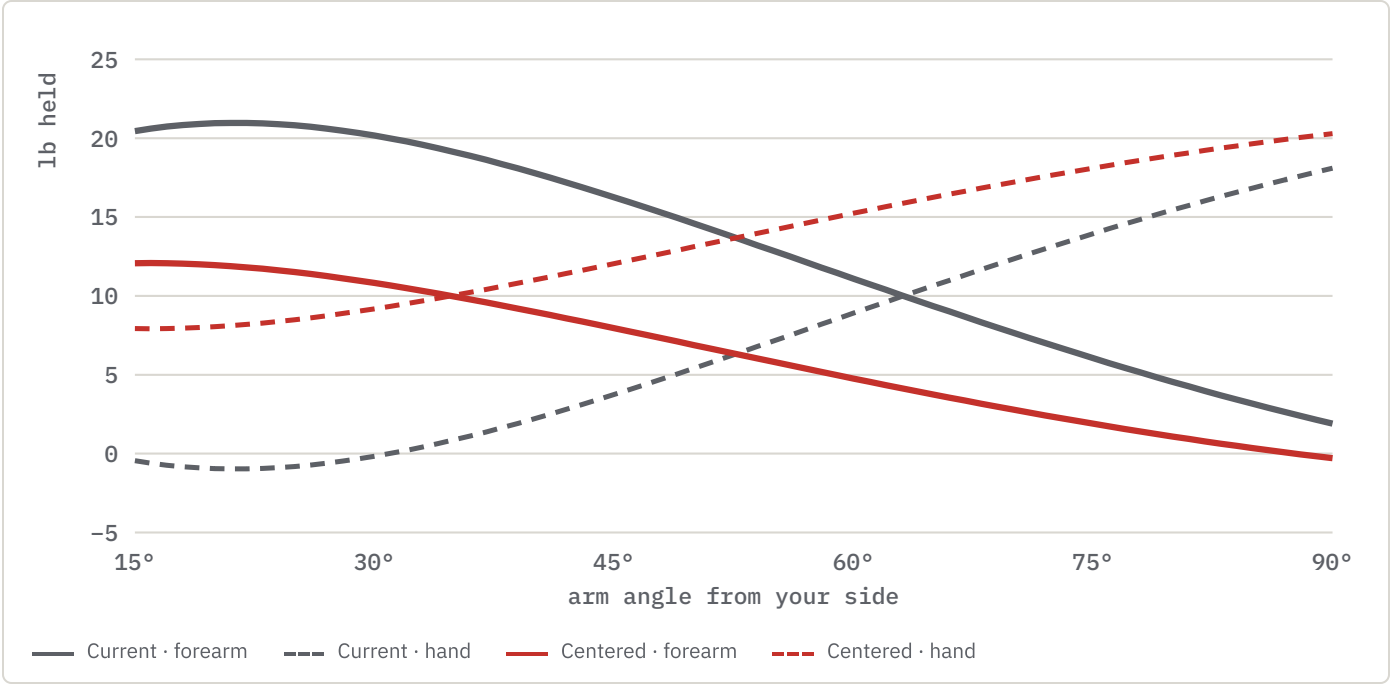
Drawn to scale from the model: true arm length, true cable-exit position, every arm position pivoting about the same shoulder. The cable passes in front of the body, so from behind it is hidden (dotted).

WHERE THE 20 LB SITS

At the hardest point of the test movement (44°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



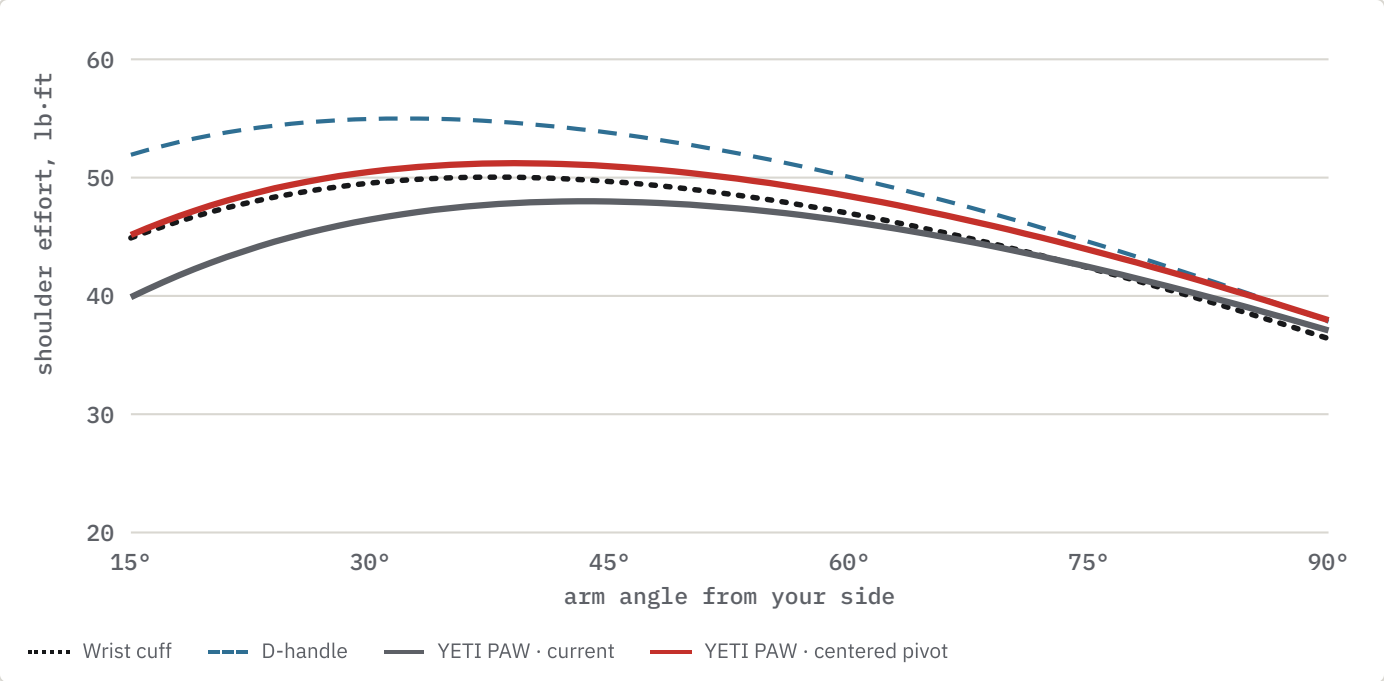
Reading the bars: red is pressure on the forearm pad, blue is load in the hand; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap. The two YETI PAW variants sit at different points between a D-handle and full forearm support.



Reading the chart: the same split through the whole movement. At every arm angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb. The current pivot (gray) is forearm-led for most of the movement and hands over to the grip near the top. The centered pivot (red) keeps hand and pad closer to even through the middle.

SAME PATH AS A CUFF

Where the cable hooks on also shapes how the resistance builds and fades through the movement. Both variants are close to a wrist cuff; the centered pivot follows it most closely.



Reading the chart: higher means more effort at that arm angle (lb·ft: pounds times feet of leverage at the shoulder). The dotted black line is a wrist cuff. The red line (centered pivot) runs alongside it the whole way, a few percent above. The gray line (current pivot) starts a little lower and meets it near the top. A plain D-handle (blue dashes) is highest at the start because the cable hangs on the very end of the arm.

Both stay

The current pivot suits lifters who want the forearm to carry the load and the grip to relax. The centered pivot suits lifters who want a balanced, D-ring or cuff-like feel. Same parts, different side plates.

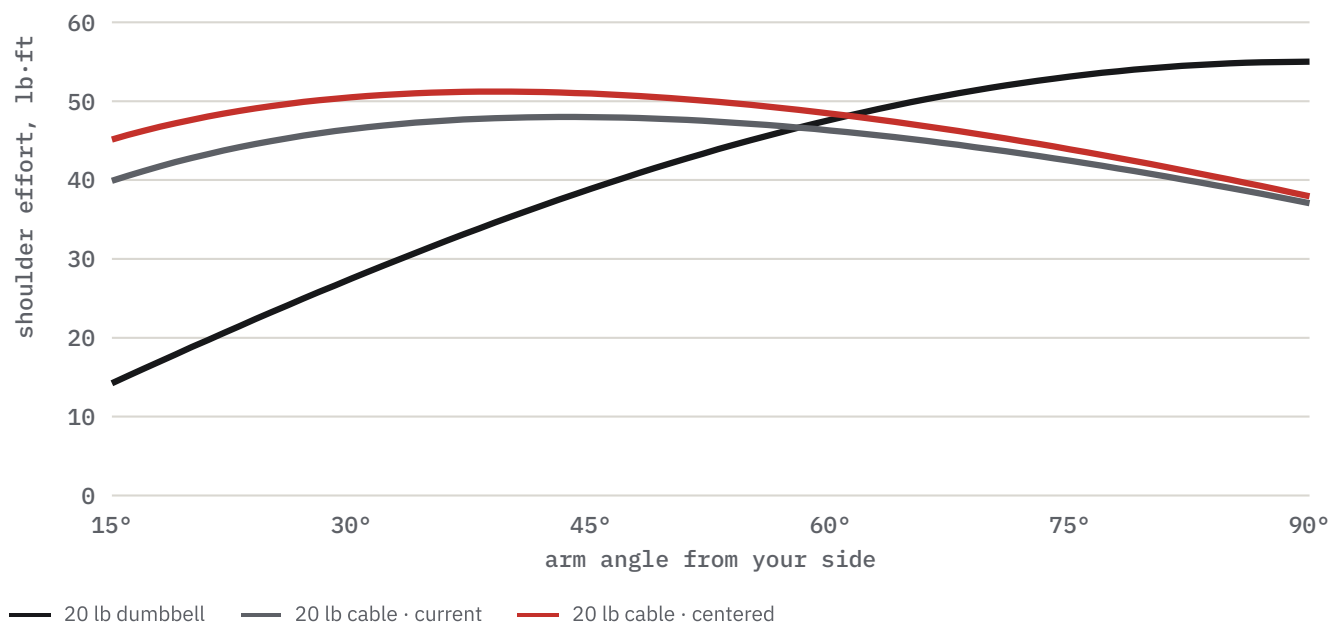
APPENDIX · EFFORT NUMBERS

ANGLE	DUMBBELL	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
15°	14	52	45	45	40
30°	27	55	50	50	46
45°	39	54	50	51	48
60°	48	50	47	48	46
75°	53	45	42	44	42
90°	55	38	36	38	37

Reading the table: shoulder effort in lb·ft (pounds times feet of leverage at the shoulder) for a 20 lb load at each arm angle of the test movement. The dumbbell column is a 20 lb dumbbell, for comparison.

APPENDIX · CABLE VERSUS DUMBBELL

For context: a cable from the floor beside the opposite foot is a different exercise from a dumbbell. A dumbbell pulls straight down, so it is easy at the bottom and hardest with the arm level. The cable pulls across the arm from the start, so it is about 3× harder at the bottom and eases off near the top. Every number in this study is for the cable.



Reading the chart: black is a 20 lb dumbbell, gray and red are a 20 lb cable on the two YETI PAW handles. Higher means the shoulder works harder.

APPENDIX · HAND AND FOREARM, IN DETAIL

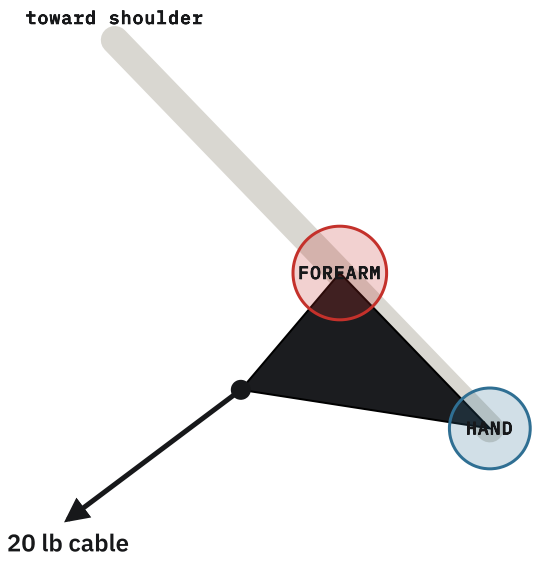
HOLDING

Hand and forearm together hold the cable. The two always add up to exactly 20 lb.

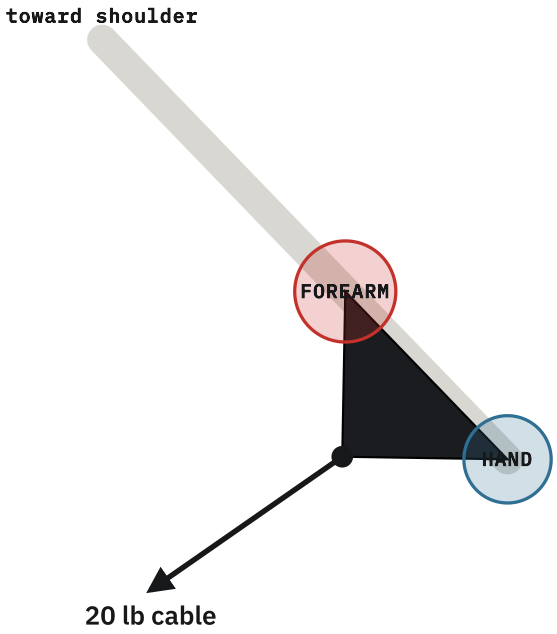
SQUEEZE

The cable hooks on to one side, so it tries to twist the handle. Hand and forearm press against each other to stop it. You feel the pressure, but it cancels out and adds nothing to the load.

CURRENT — PIVOT BELOW THE WRIST PAD, 44°



CENTERED — PIVOT BETWEEN HAND AND PAD, 44°



ANGLE	CURRENT · FOREARM / HAND	SQUEEZE	CENTERED · FOREARM / HAND	SQUEEZE
15°	20 / lifts off	10	12 / 8	5
30°	20 / 0	3	11 / 9	1
45°	16 / 4	2	8 / 12	1
60°	11 / 9	4	5 / 15	2
75°	6 / 14	4	2 / 18	1
90°	2 / 18	2	lifts off / 20	0

Reading the table: pounds of the 20 lb cable at each arm angle. “Lifts off” means that contact stops touching, so the other one holds all 20.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Cable exit 8″ off the floor (a Voltra on a 45° rack bracket), about 10″ out from the outside of the left leg
- Fist center 29″ off the floor, arm hanging
- Top of the outside of the delt 64″ off the floor
- Handle holes 120 / 140 / 85 mm (checked against a photo)

ASSUMED

- Shoulder joint center 2″ below the delt top, 8″ out from the midline
- Outside of the leg 6″ from the midline at cable-exit height, so the cable exit sits 16″ left of center
- Straight arm, 33″ shoulder to fist
- Wrist cuff ring 3″ behind the fist center, 1.2″ below the wrist line
- Raise in the plane of the body, seen from behind
- Pad presses square to the forearm; the grip does the rest

CHECKS

- Shoulder effort = force × its distance from the shoulder joint; the cable pulls from the yoke pivot toward the cable exit
- Re-derived independently from scratch; every number matched within 1%
- Moving the cuff ring anywhere from 2.5–3.5″ back and 0.8–1.5″ down: the centered pivot stays within 6% of the cuff
- Moving the shoulder or cable-exit guesses by several inches: the forearm load at the hardest point stays 15–18 lb (current) vs 7–9 lb (centered)

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YETI PAW · CABLE CURL · FACING THE VOLTRA · DESIGN STUDY

BICEP CURL: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the bicep curl. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load more evenly between hand and wrist, closer to a D-ring handle or a wrist cuff. Both are offered as variants.

THE BOTTOM LINE

13 + 7 lb

Current: forearm-led. At the hardest point of the bicep curl (115°) the forearm pad carries 13 lb of a 20 lb cable and the hand 7 lb.

6 + 14 lb

Centered: balanced. The same moment: 6 lb on the pad, 14 lb in the hand. A plain D-handle puts all 20 in the hand.

+16%

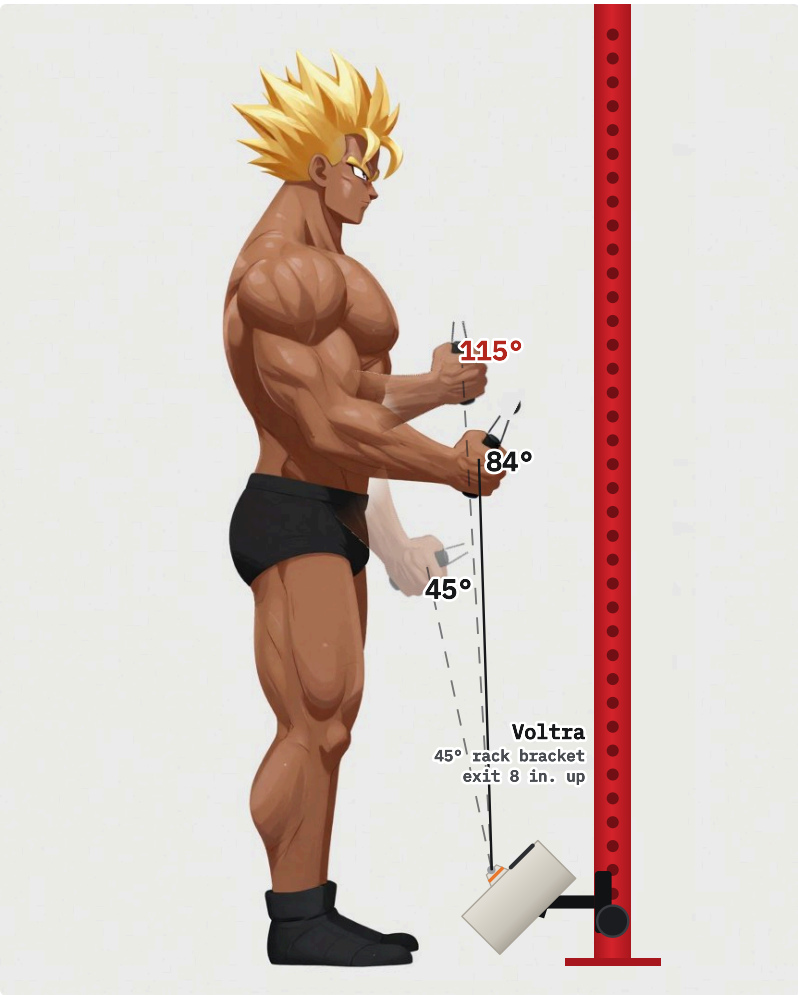
A little more work. For the same weight, the centered pivot's peak elbow (biceps) effort is 16% higher than the current pivot's; a wrist cuff sits in between (+8% vs current). The movement and its shape stay the same.

How sure are we? Confident on the direction, not yet on feel. The numbers come from the same physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the stance and forearm length assumed (see the appendix). Moving those guesses changes the exact pounds, not the pattern: across the cases checked, the forearm carries 12–14 lb at the hardest point with the current pivot and 5–6 lb with the centered one.

Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

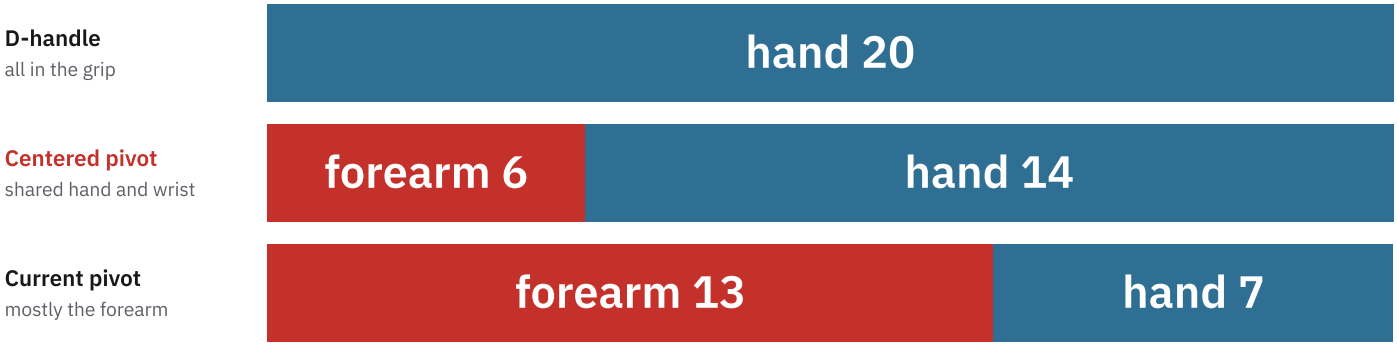
A standing cable curl, seen from the side, facing a Voltra on a 45° rack bracket: its cable exits 8 in. off the floor, about 16 in. in front of the elbow. The elbow stays at the side while the forearm rises. Early in the curl the cable runs nearly along the forearm and helps rather than resists: the YETI PAW starts carrying load at 42° (current pivot) or 35° (centered). The sheet follows the curl from 30° to 140°, measured from the arm hanging straight.



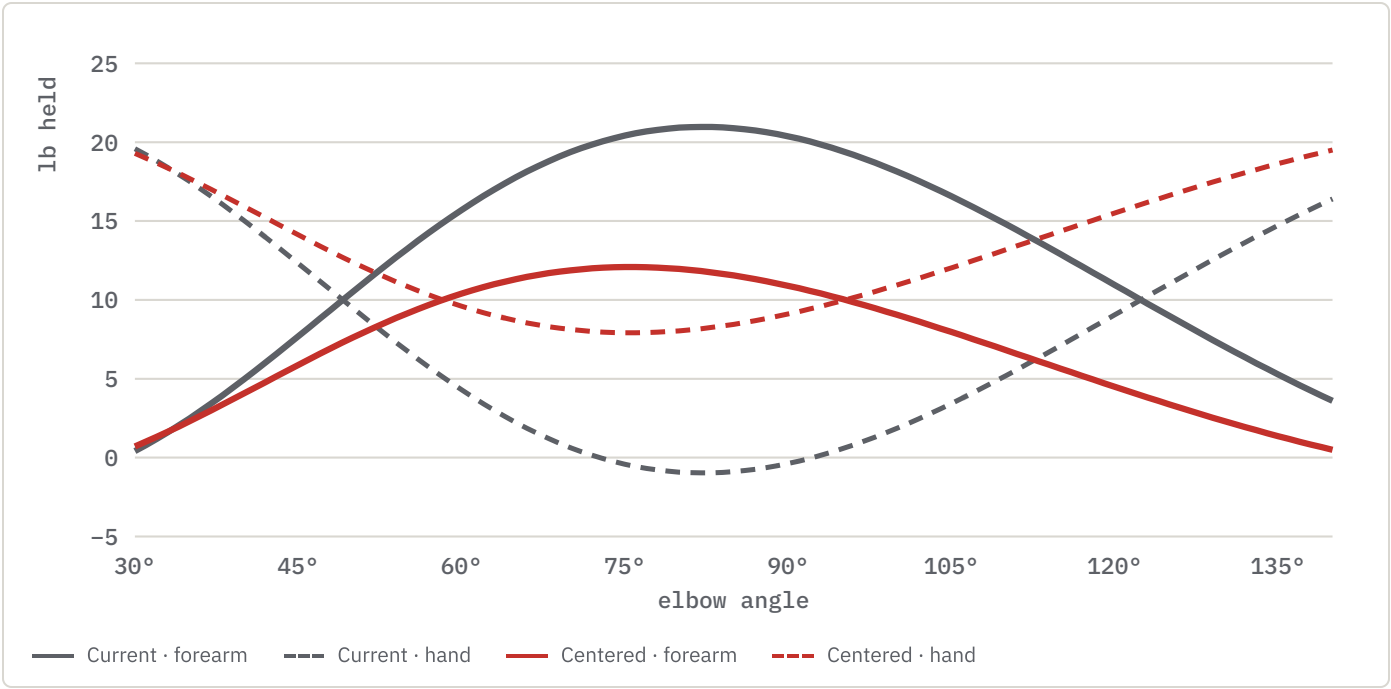
Drawn to the model: one forearm, one length, pivoting about one elbow at the exact angles shown (solid = the pose as drawn, faded = other points in the curl), and the Voltra's cable exit where the model puts it, 8 in. off the floor he stands on. The red angle is the hardest point.

WHERE THE 20 LB SITS

At the hardest point (115°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



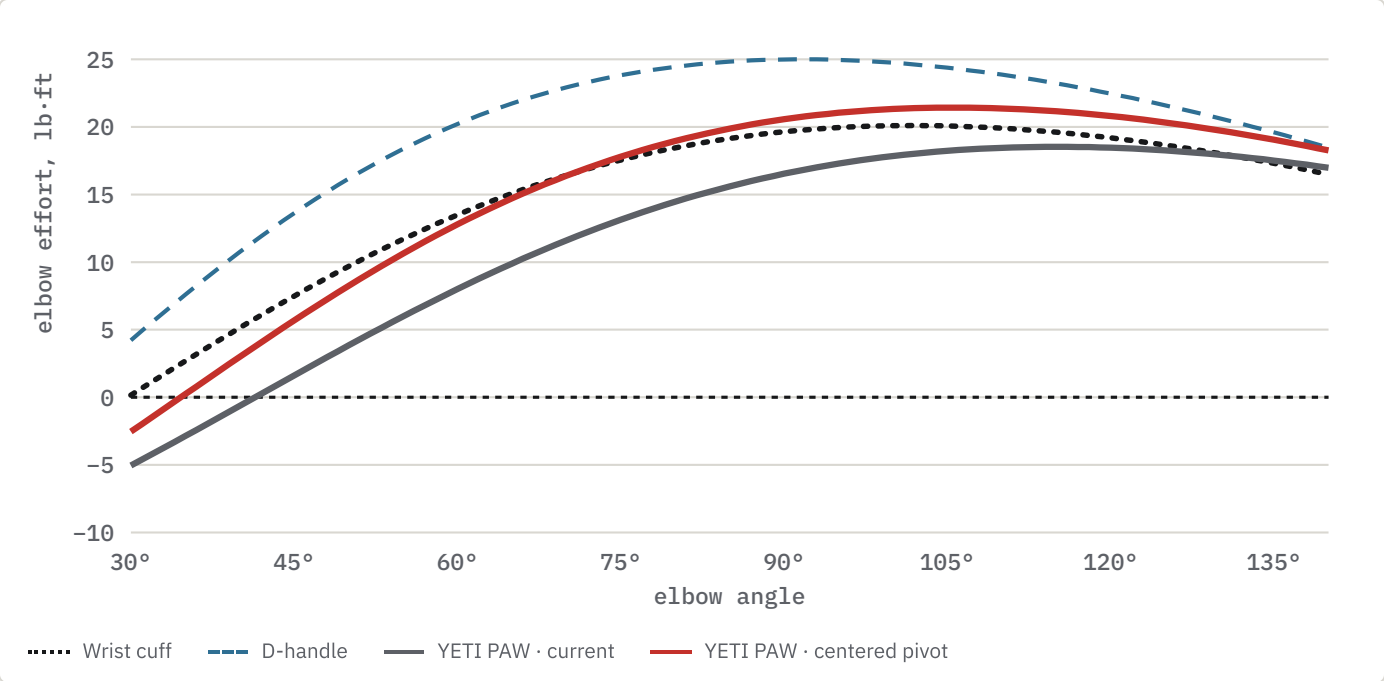
Reading the bars: red is pressure on the forearm pad, blue is load in the hand; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

SAME PATH AS A CUFF

Where the cable hooks on also shapes how the resistance builds and fades. Both variants stay close to a wrist cuff; the centered pivot follows it more closely (within 11% of the peak, against 23% for the current pivot).



Reading the chart: higher means more elbow effort at that angle (lb·ft: pounds times feet of leverage at the elbow). Below zero, the cable is pulling the limb along rather than resisting it.

Both stay

The current pivot suits lifters who want the forearm to carry the load and the grip to relax. The centered pivot suits lifters who want a balanced, D-ring or cuff-like feel. Same parts, different side plates.

APPENDIX · EFFORT NUMBERS

ANGLE	DUMBBELL	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
30°	12	4	0	−3	−5
45°	18	14	7	6	2
60°	22	20	13	13	8
75°	24	24	18	18	13
90°	25	25	20	21	17
105°	24	24	20	21	18
120°	22	22	19	21	18
135°	18	20	17	19	18

Reading the table: elbow (biceps) effort in lb·ft for a 20 lb load at each angle. Negative means the cable is helping the movement at that angle. The dumbbell column is a 20 lb dumbbell, for comparison.

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · FOREARM / HAND	SQUEEZE	CENTERED · FOREARM / HAND	SQUEEZE
30°	0 / 20	15	1 / 19	11
45°	8 / 12	19	6 / 14	12
60°	16 / 4	17	10 / 10	10
75°	20 / lifts off	11	12 / 8	5
90°	20 / lifts off	3	11 / 9	1
105°	17 / 3	2	8 / 12	1
120°	11 / 9	5	4 / 16	2
135°	5 / 15	4	1 / 19	1

Reading the table: pounds of the 20 lb cable. “Lifts off” means that contact stops touching, so the other one holds all 20. Squeeze is the pinch from the cable trying to twist the handle; it cancels out and adds no load.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Elbow to fist center 15″
- Voltra on a 45° rack bracket, cable exit 8″ up, 16″ in front of the elbow; elbow pinned at the side, 44″ up
- Movement in one plane (side view, facing the Voltra)
- Pad presses square to the forearm on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Effort = force × its distance from the elbow; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: forearm length (13.5–16.5″) and stance (12–20″ from the elbow) moved; forearm load at the hardest point stays 12–14 lb (current) vs 5–6 lb (centered)

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YETI PAW · CABLE CURL · FACING AWAY FROM THE VOLTRA · DESIGN STUDY

BAYESIAN CURL: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the Bayesian curl. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load more evenly between hand and wrist, closer to a D-ring handle or a wrist cuff. Both are offered as variants.

THE BOTTOM LINE

13 + 7 lb

Current: forearm-led. At the hardest point of the Bayesian curl (89°) the forearm pad carries 13 lb of a 20 lb cable and the hand 7 lb.

6 + 14 lb

Centered: balanced. The same moment: 6 lb on the pad, 14 lb in the hand. A plain D-handle puts all 20 in the hand.

+16%

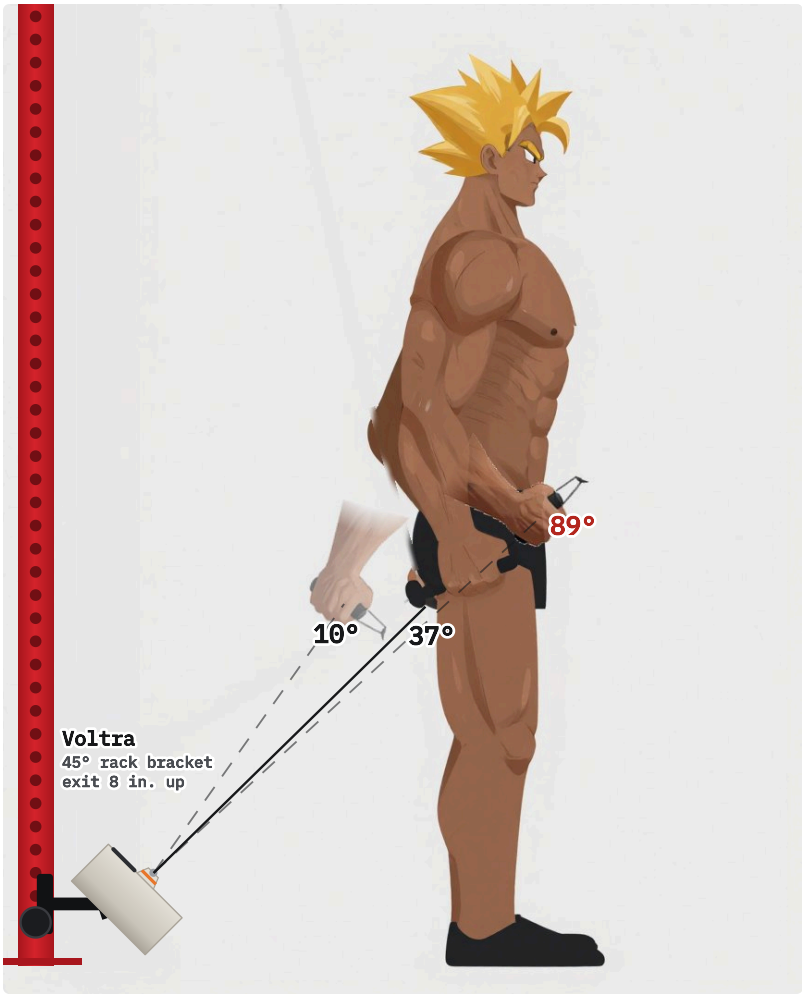
A little more work. For the same weight, the centered pivot's peak elbow (biceps) effort is 16% higher than the current pivot's; a wrist cuff sits in between (+8% vs current). The movement and its shape stay the same.

How sure are we? Confident on the direction, not yet on feel. The numbers come from the same physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the stance and forearm length assumed (see the appendix). Moving those guesses changes the exact pounds, not the pattern: across the cases checked, the forearm carries 12–14 lb at the hardest point with the current pivot and 5–6 lb with the centered one.

Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

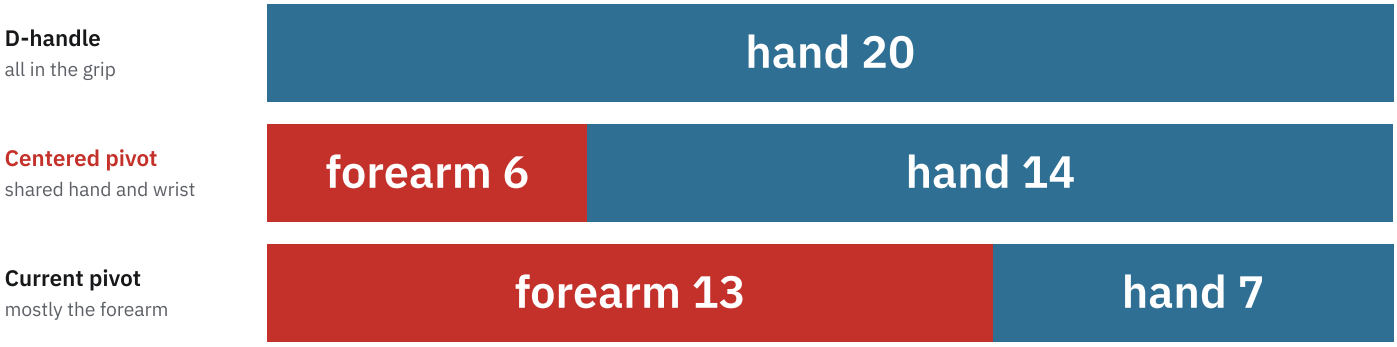
A Bayesian cable curl, seen from the side: facing away from the Voltra, with the upper arm held about 25° behind the body. The cable comes from a Voltra on a 45° rack bracket behind him: it exits 8 in. off the floor, about 28 in. behind the shoulder. The forearm curls from straight (0°) to 130°.



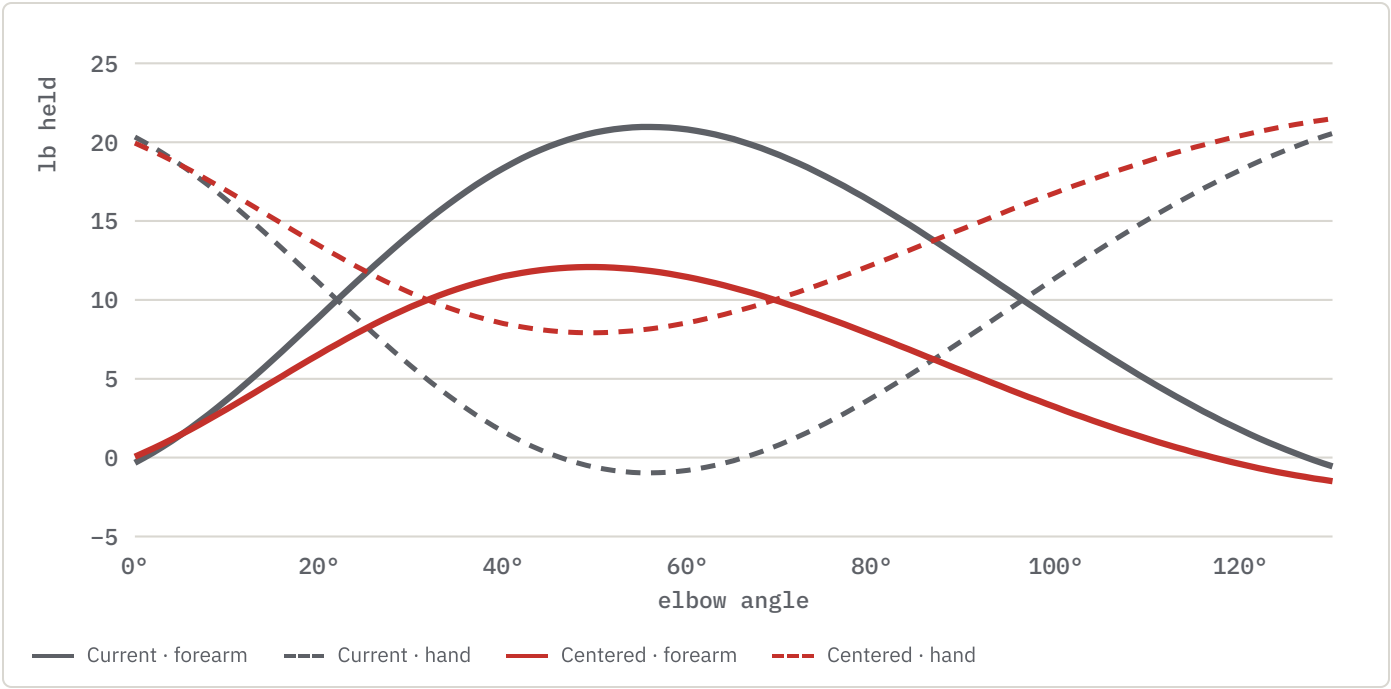
Drawn to the model: one forearm, one length, pivoting about one elbow at the exact angles shown (solid = the pose as drawn, faded = other points in the curl), and the Voltra's cable exit where the model puts it, 8 in. off the floor he stands on. The red angle is the hardest point.

WHERE THE 20 LB SITS

At the hardest point (89°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



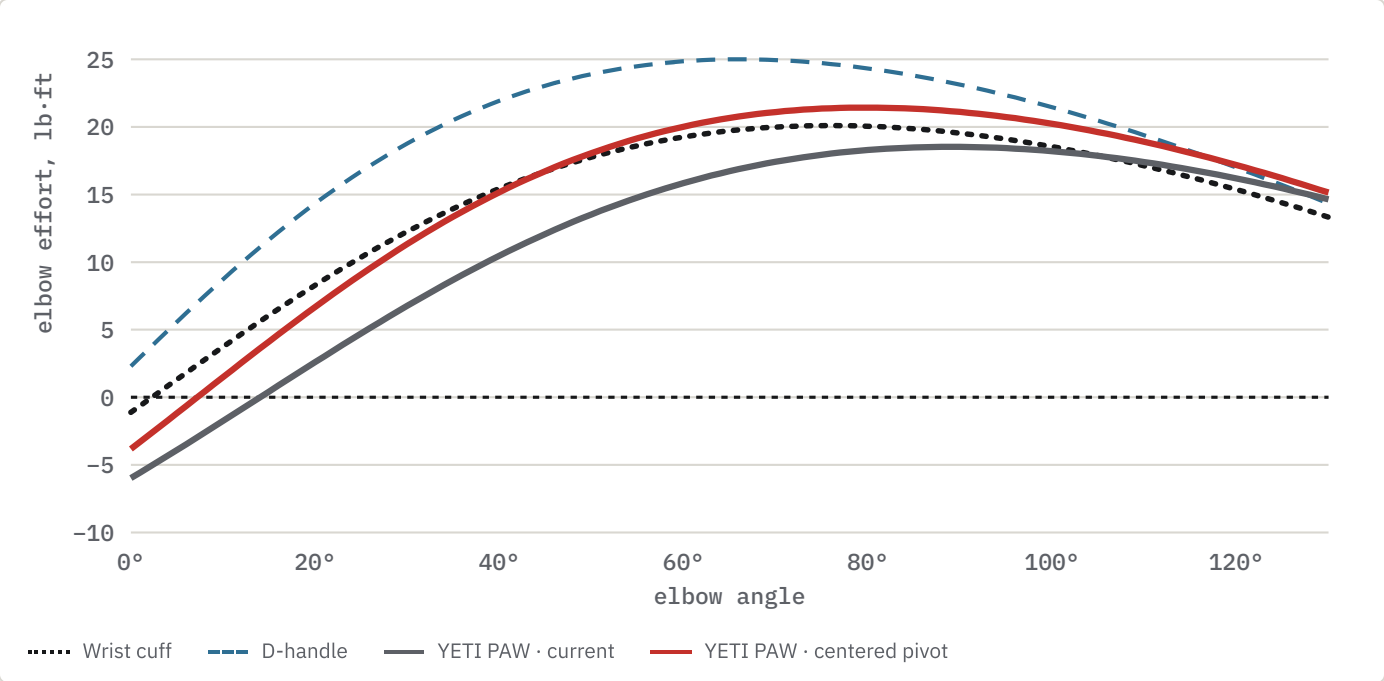
Reading the bars: red is pressure on the forearm pad, blue is load in the hand; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

SAME PATH AS A CUFF

Where the cable hooks on also shapes how the resistance builds and fades. Both variants stay close to a wrist cuff; the centered pivot follows it more closely (within 11% of the peak, against 23% for the current pivot).



Reading the chart: higher means more elbow effort at that angle (lb·ft: pounds times feet of leverage at the elbow). Below zero, the cable is pulling the limb along rather than resisting it.

Both stay

The current pivot suits lifters who want the forearm to carry the load and the grip to relax. The centered pivot suits lifters who want a balanced, D-ring or cuff-like feel. Same parts, different side plates.

APPENDIX · EFFORT NUMBERS

ANGLE	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
0°	2	−1	−4	−6
15°	12	6	4	0
30°	19	12	11	7
45°	23	17	17	12
60°	25	19	20	16
75°	25	20	21	18
90°	23	20	21	19
105°	20	18	20	18
120°	17	15	17	16

Reading the table: elbow (biceps) effort in lb·ft for a 20 lb load at each angle. Negative means the cable is helping the movement at that angle.

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · FOREARM / HAND	SQUEEZE	CENTERED · FOREARM / HAND	SQUEEZE
0°	lifts off / 20	14	0 / 20	10
15°	6 / 14	18	5 / 15	12
30°	14 / 6	18	10 / 10	10
45°	20 / 0	12	12 / 8	6
60°	20 / lifts off	5	11 / 9	2
75°	18 / 2	1	9 / 11	1
90°	13 / 7	4	5 / 15	2
105°	7 / 13	4	2 / 18	1
120°	2 / 18	2	lifts off / 20	0

Reading the table: pounds of the 20 lb cable. “Lifts off” means that contact stops touching, so the other one holds all 20. Squeeze is the pinch from the cable trying to twist the handle; it cancels out and adds no load.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Elbow to fist center 15″
- Voltra on a 45° rack bracket, cable exit 8″ up, 28″ behind the shoulder; upper arm held 25° behind vertical
- Movement in one plane (side view, facing away from the Voltra)
- Pad presses square to the forearm on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Effort = force × its distance from the elbow; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: forearm length (13.5–16.5″), upper-arm angle (15–35° back) and stance (20–36″ behind) moved; forearm load at the hardest point stays 12–14 lb (current) vs 5–6 lb (centered)

Designed by Schnibbz · YETI STRENGTH · YETI PAW research v2.0 · 2026-10-17

YETI PAW · CABLE FLY · PAIR OF CABLES · DESIGN STUDY

CABLE CHEST FLY: TWO WAYS TO CARRY THE LOAD

The same comparison as the lateral raise, on the cable chest fly. The current pivot puts most of the load on the forearm pad. The centered pivot hooks on near the wrist and shares the load more evenly between hand and wrist, closer to a D-ring handle or a wrist cuff. Both are offered as variants.

THE BOTTOM LINE

17 + 3 lb

Current: forearm-led. At the hardest point of the cable chest fly (31°) the forearm pad carries 17 lb of a 20 lb cable and the hand 3 lb.

8 + 12 lb

Centered: balanced. The same moment: 8 lb on the pad, 12 lb in the hand. A plain D-handle puts all 20 in the hand.

+7%

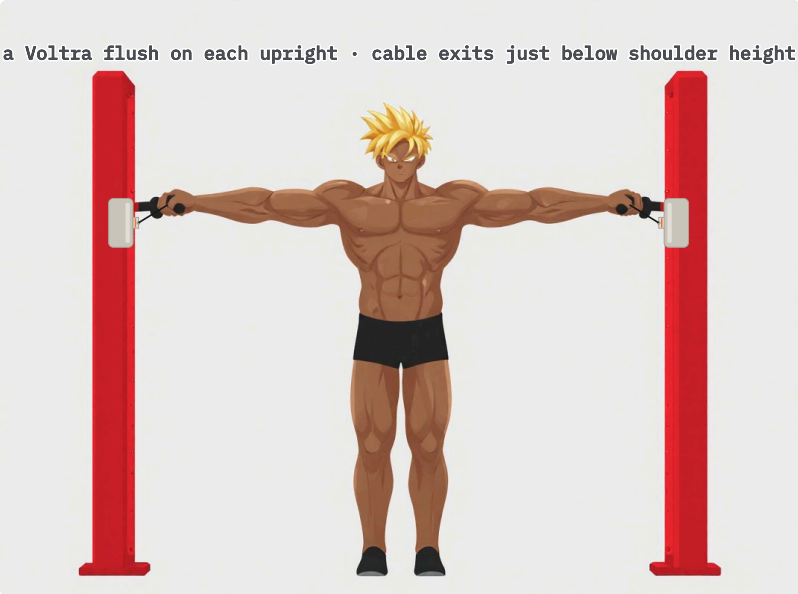
A little more work. For the same weight, the centered pivot's peak shoulder (chest) effort is 7% higher than the current pivot's; a wrist cuff sits in between (+4% vs current). The movement and its shape stay the same.

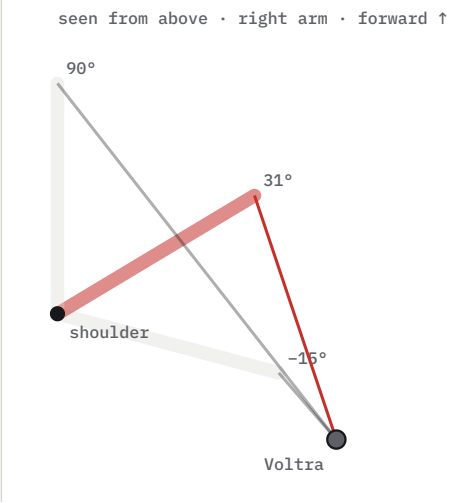
How sure are we? Confident on the direction, not yet on feel. The numbers come from the same physics model as the lateral raise study, built on one real lifter (6′5″, measured), with the cable-exit position assumed (see the appendix). Moving those guesses changes the exact pounds, not the pattern: across the cases checked, the forearm carries 17 lb at the hardest point with the current pivot and 8 lb with the centered one.

Wear the handle so the pad sits on the side of the forearm away from the cable, as on the lateral raise.

THE MOVEMENT

A standing cable chest fly between a pair of Voltras flush-mounted on uprights, cable exits just below shoulder height, about 4 ft out to each side and 18 in. behind the shoulders. Each arm works the same way, so the sheet follows one arm, seen from above, as it sweeps from slightly behind the body (−15°) to pointing straight ahead (90°).

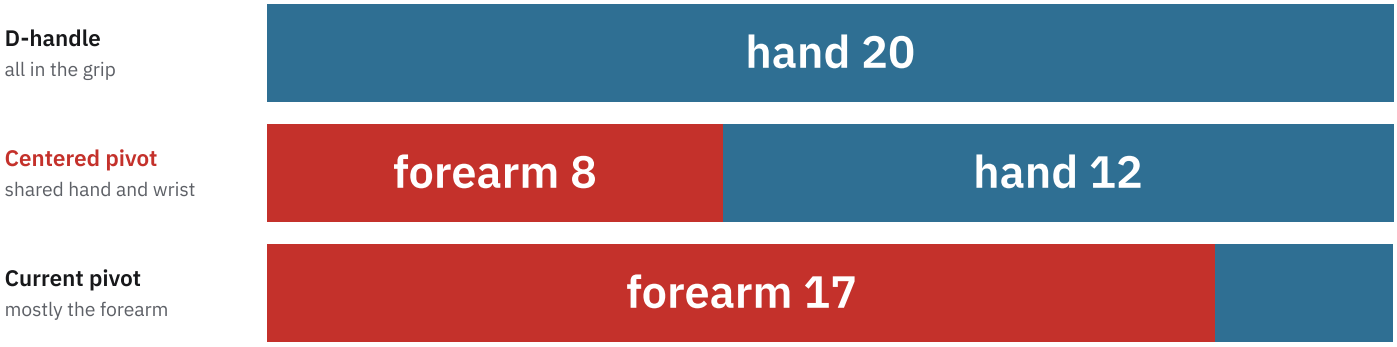




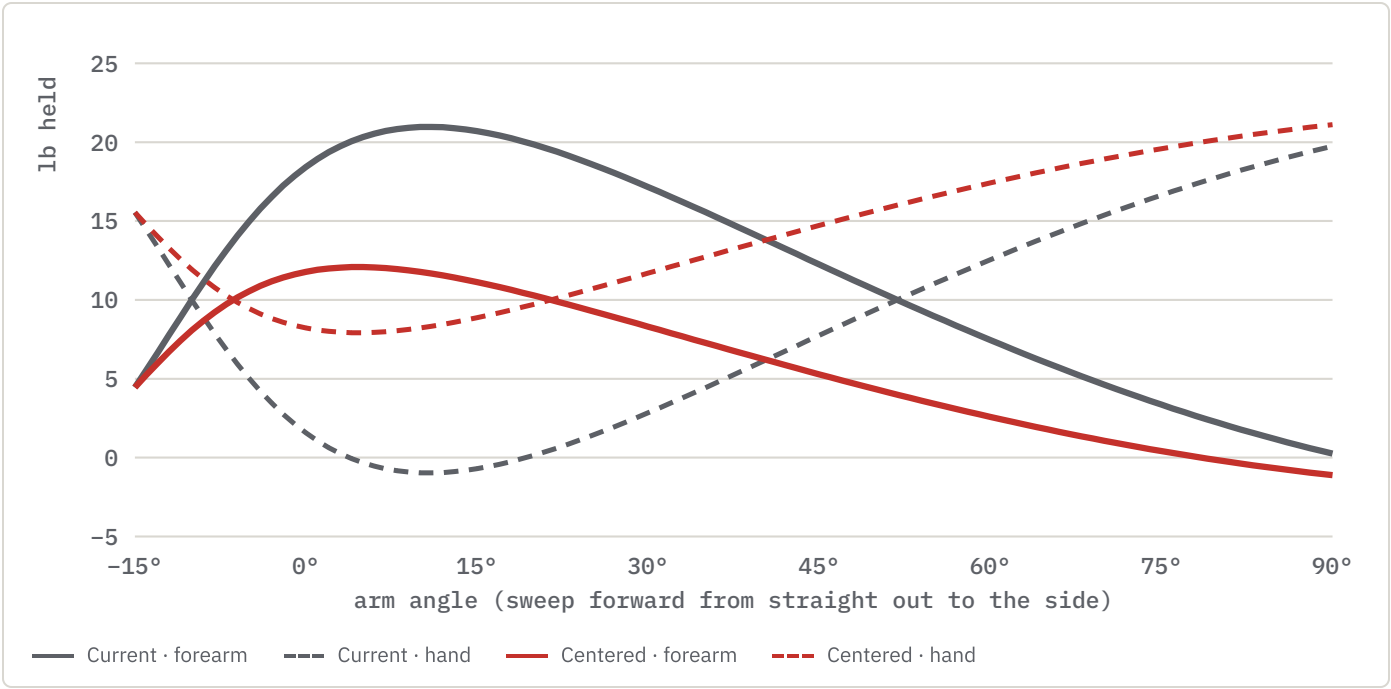
From the front: both cables come from Voltras flush-mounted on the uprights, just below shoulder height. The sweep happens in depth, so the inset shows it from above, drawn from the model: −15° (stretched, start), the hardest point, and 90° (arms straight ahead).

WHERE THE 20 LB SITS

At the hardest point (31°), here is how a 20 lb cable is carried, with a plain D-handle for reference.



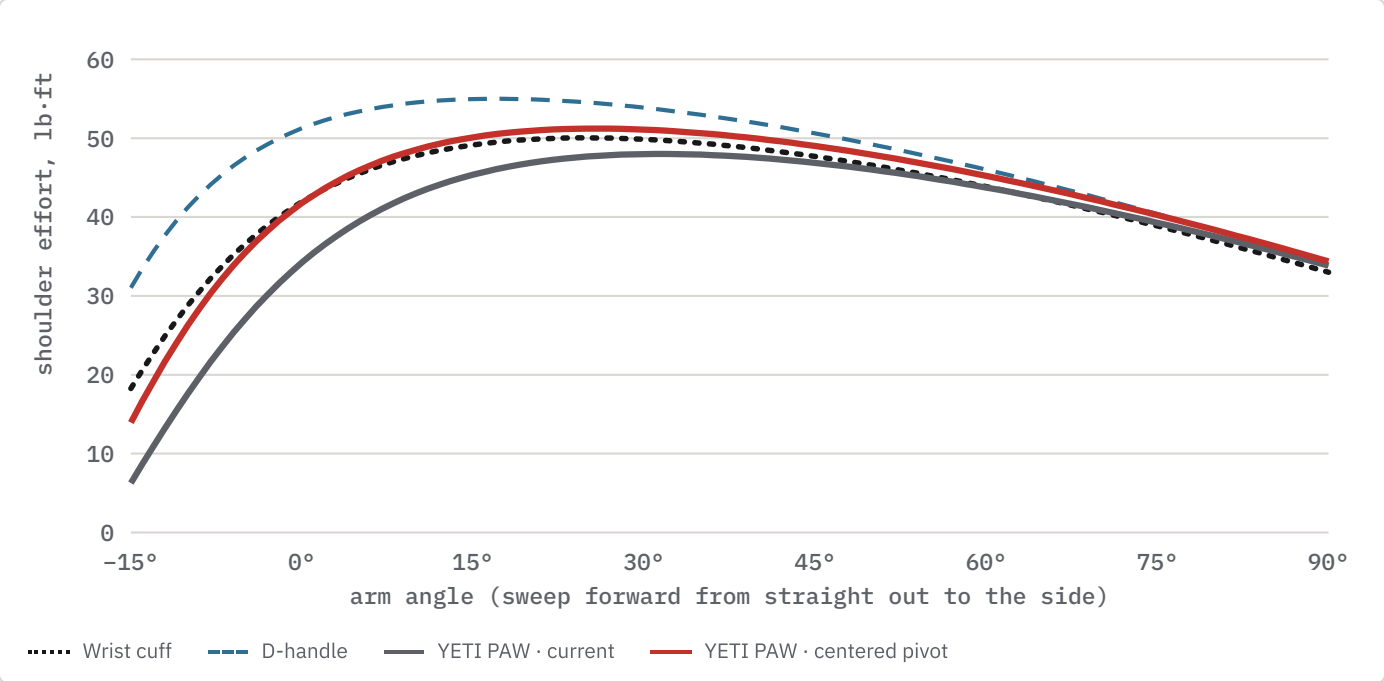
Reading the bars: red is pressure on the forearm pad, blue is load in the hand; each bar adds up to the 20 lb cable. A wrist cuff puts everything on the wrist strap.



Reading the chart: the same split through the whole movement. At every angle the solid line (forearm pad) and the dashed line of the same color (hand) add up to 20 lb.

SAME PATH AS A CUFF

Where the cable hooks on also shapes how the resistance builds and fades. Both variants stay close to a wrist cuff; the centered pivot follows it more closely (within 8% of the peak, against 22% for the current pivot).



Reading the chart: higher means more shoulder effort at that angle (lb·ft: pounds times feet of leverage at the shoulder).

Both stay

The current pivot suits lifters who want the forearm to carry the load and the grip to relax. The centered pivot suits lifters who want a balanced, D-ring or cuff-like feel. Same parts, different side plates.

APPENDIX · EFFORT NUMBERS

ANGLE	D-HANDLE	WRIST CUFF	CENTERED	CURRENT
−15°	31	18	14	6
0°	51	42	42	34
15°	55	49	50	45
30°	54	50	51	48
45°	51	48	49	47
60°	46	44	45	44
75°	40	39	40	39
90°	34	33	34	34

Reading the table: shoulder (chest) effort in lb·ft for a 20 lb load at each angle.

APPENDIX · HAND AND FOREARM, IN DETAIL

ANGLE	CURRENT · FOREARM / HAND	SQUEEZE	CENTERED · FOREARM / HAND	SQUEEZE
−15°	4 / 16	18	4 / 16	12
0°	18 / 2	14	12 / 8	7
15°	20 / lifts off	5	11 / 9	1
30°	17 / 3	2	8 / 12	1
45°	12 / 8	4	5 / 15	2
60°	7 / 13	4	3 / 17	2
75°	3 / 17	3	0 / 20	0
90°	0 / 20	0	lifts off / 20	1

Reading the table: pounds of the 20 lb cable. “Lifts off” means that contact stops touching, so the other one holds all 20. Squeeze is the pinch from the cable trying to twist the handle; it cancels out and adds no load.

APPENDIX · HOW THESE NUMBERS WERE MADE

MEASURED

- Lifter 6′5″
- Shoulder joint 62″ off the floor, fist center 29″ with the arm hanging (33″ shoulder to fist)
- Handle holes: current 120 / 140 / 85 mm, centered 120 / 85 / 85 mm

ASSUMED

- Straight arm, 33″ shoulder to fist
- Voltras flush-mounted, cable exits just below shoulder height, 48″ out from the midline and 18″ behind the shoulders; straight arm
- Movement in one plane (seen from above, one arm of the pair)
- Pad presses square to the forearm on the side away from the cable; the grip does the rest
- Wrist cuff ring 3″ behind the fist center

CHECKS

- Same statics as the lateral raise study (re-derived independently there; matched within 1%)
- Effort = force × its distance from the shoulder; the cable pulls from the yoke pivot toward the cable exit
- Sensitivity: cable-exit position (40–56″ out, 8–28″ behind the shoulders) and arm length (30″) moved; forearm load at the hardest point stays 17 lb (current) vs 8 lb (centered)

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YETI PAW · FROM STUDY TO HARDWARE

THE CENTERED PLATE, CHECKED ON EVERY MOVEMENT

The study left two things open on the hardware side: the plate outline and the yoke's swing clearance. Both are done, along with a better carabiner eye.

V2.0 · 2026-10-17

SIDE PLATE

4.4 / 1.5

Original plate. Static / fatigue safety factor at a 100 lb cable, ELEGOO PETG-CF.

4.4 / 1.6

Centered plate. Yoke hole at the isosceles apex, 85 mm from handle and pad; the yoke-to-handle side swept R20.

R8-2RS

Same parts. Same handle, pad, yoke and hardware; plain R8-2RS handle bearings.

YOKE SWING

The yoke follows the cable (plus the 5.4° eye offset). Angles from the yoke's rest line, + toward the handle.

MOVEMENT	YOKE NEEDS	PLATE ALLOWS	COVERED
Lateral raise	−41° … +29°	−81° … +103°	yes
Bicep curl	−37° … +92°	−81° … +103°	yes
Bayesian curl	−50° … +95°	−81° … +103°	yes
Cable chest fly	−47° … +75°	−81° … +103°	yes

CARABINER EYE

One plain bronze sleeve, flush both faces. McMaster 6391K256, oil-embedded SAE 841, 3/4″ bore × 7/8″ OD × 1/2″ long, pressed through the 1/2″ eye, so both rims the carabiner bears on are bronze.

Contact model: bronze only up to 40° of tilt; the worst tilt in use is about 5° (the 5.4° eye offset).

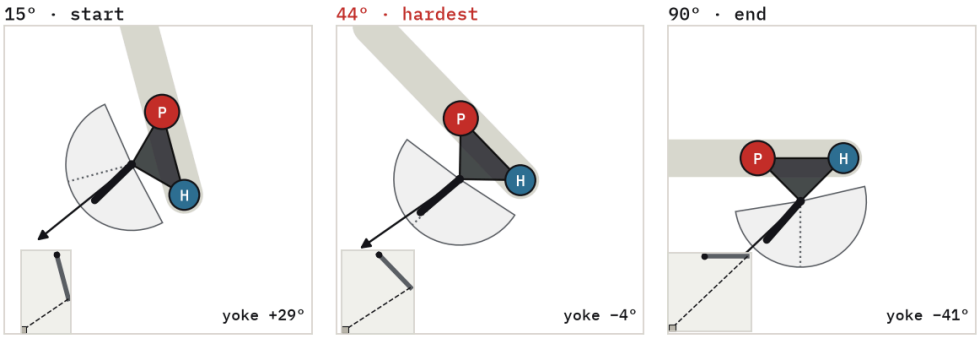
ORIENTATION CHECK

Each movement in its own view at the start, the hardest point and the end, drawn from the same model as the numbers: where the handle, pad and pivot sit, and where the cable points the yoke inside its free swing. The small inset shows the whole limb and the Voltra.



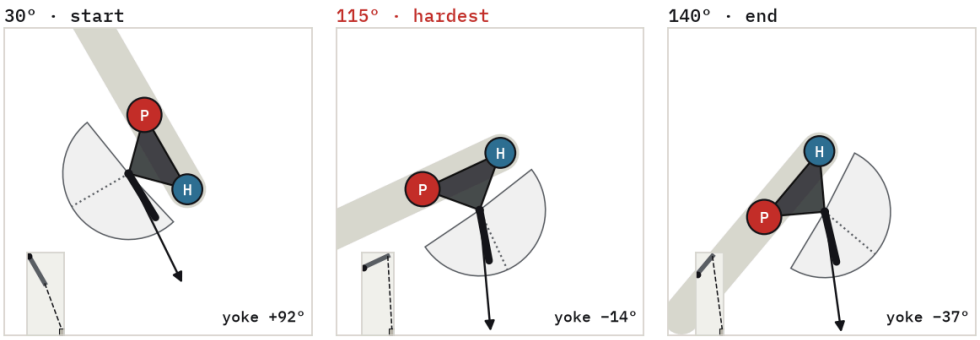
LATERAL RAISE

seen from behind

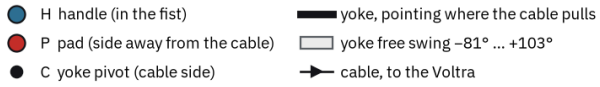


BICEP CURL

seen from his right side

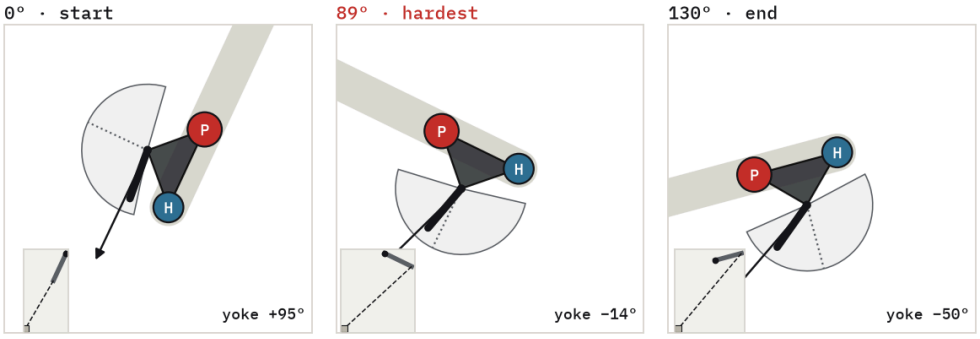


ORIENTATION CHECK, CONTINUED



BAYESIAN CURL

seen from his right side



CABLE CHEST FLY

seen from above

